

# Vector-space containers (VCont)

A self-contained treatment: the biproduct collapse, algebraic structure,  
and the admissibility frontier

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*Draft — work in progress; not yet neighbour-reviewed. Subject matter prepared for  
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*Reading guide and honesty note.* Every theorem below is tagged in prose with its epistemic grade: **proved** (a complete pen-and-paper proof exists in the cited internal file), **lean-verified** (machine-checked, sorry-free, in Lean 4), or **OPEN** (a precisely stated conjecture that is *not* settled). The headline result of §2 is proved and lean-verified. The frontier question of §4 (“Question (Q)”) is **OPEN** and is conjectured independent of ZFC; it is presented as such and nowhere treated as settled. This note has not yet had a neighbour review.

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Fix a field  $k$ . A *container* in a category  $\mathcal{C}$  is a pair (shapes, positions): a set  $S$  together with a position object  $P_s \in \mathcal{C}$  for each  $s \in S$ , extending to an endofunctor  $\llbracket S, P \rrbracket$  of  $\mathcal{C}$ . Over  $\mathcal{C} = \mathbf{Set}$  this is the classical theory of Abbott, Altenkirch and Ghani: containers are precisely the polynomial functors, and the extension functor is a full and faithful embedding. This note asks what survives when the positions are *vector spaces* — the base  $\mathcal{C} = \mathbf{Vec}_{\text{fd}}$  of finite-dimensional  $k$ -vector spaces — and reports the answers the programme has established. The one-line summary: *the coproduct/product distinction that makes **Set** containers rigid becomes a biproduct over **Vec**, and much of what the **Set** theory reads off “for free” collapses.*

## 1 Definition of VCont and its extension functor

Write  $\mathbf{Vec} = \mathbf{Vec}_k$  for all  $k$ -vector spaces,  $\mathbf{Vec}_{\text{fd}}$  for the finite-dimensional ones, and  $\mathbf{Vec}(P, W)$  for the hom vector space. All functors below are **Vec**-functors: their action on homs  $\mathbf{Vec}(V, W) \rightarrow \mathbf{Vec}(FV, FW)$  is  $k$ -linear.

**Definition 1** (linear / vector-space container). A *vector-space container* (object of  $\mathbf{VCont} := \text{Fam}(\mathbf{Vec}^{\text{op}})$ , or of  $\text{Fam}(\mathbf{Vec}_{\text{fd}}^{\text{op}})$  when finiteness is imposed) is a pair  $(S, (P_s)_{s \in S})$  with  $S$  a set (the *shapes*) and each  $P_s \in \mathbf{Vec}$  a *position space*. A morphism  $(S, (P_s)) \rightarrow (T, (Q_t))$  is a pair  $(f, (\varphi_s))$  with  $f: S \rightarrow T$  a function and, for each  $s \in S$ , a linear map  $\varphi_s: Q_{f(s)} \rightarrow P_s$  — *covariant on shapes, contravariant on positions*, exactly as for **Set** containers. Write  $n_s := \dim P_s$ .

**Definition 2** (extension). The *extension* of  $(S, P)$  is the functor  $\llbracket S, P \rrbracket: \mathbf{Vec} \rightarrow \mathbf{Vec}$ ,

$$\llbracket S, P \rrbracket W := \bigoplus_{s \in S} \mathbf{Vec}(P_s, W).$$

Choosing a basis of  $P_s$  gives  $\mathbf{Vec}(P_s, W) \cong W^{n_s}$  naturally in  $W$ , so  $\llbracket S, P \rrbracket W \cong \bigoplus_{s \in S} W^{\oplus n_s}$ , a “direct sum of powers”. On a morphism  $(f, (\varphi_s))$  the natural transformation is  $(g_s)_s \mapsto (g_{f(s)} \circ \varphi_s)_s$ . Writing  $h_P := \mathbf{Vec}(P, -)$  for the corepresentable at  $P$ , we have  $\llbracket S, P \rrbracket = \bigoplus_{s \in S} h_{P_s}$  and  $h_k = \mathbf{Vec}(k, -) \cong \text{Id}$ .

Two features of **Vec** shape everything that follows.

*Coproduct = product = biproduct.* A finite coproduct in **Vec** is a biproduct, hence *equals* the product. Thus although the outer sum  $\bigoplus_{s \in S}$  in Definition 2 is written as a coproduct (a “ $\Sigma$  over shapes” of the hom-spaces  $\mathbf{Vec}(P_s, X)$ ), for finite  $S$  its underlying space is equally the product  $\prod_{s \in S} \mathbf{Vec}(P_s, X)$  — a “ $\Pi$  over shapes”. This  $\Sigma \rightsquigarrow \Pi$  coincidence is the *biproduct collapse*, and it is the single change from the set-level extension  $\llbracket S, P \rrbracket X = \coprod_{s \in S} X^{P_s}$  (a genuine disjoint union). It is made fully explicit in the Lean formalisation of §5, where the linear extension is literally  $\mathbf{LExt}(S, \mathbf{Pos}) X = (s : S) \rightarrow (\mathbf{Pos} s \rightarrow X)$ .

*Finite positions distribute over  $\oplus$ .* If  $\dim P < \infty$  then  $\mathbf{Vec}(P, \bigoplus_t V_t) \cong \bigoplus_t \mathbf{Vec}(P, V_t)$  naturally; equivalently,  $h_P$  preserves coproducts iff  $P$  is finite-dimensional (a finite-support argument on a basis of  $P$ ). This is why we usually restrict positions to  $\mathbf{Vec}_{\text{fd}}$ ; the infinite-dimensional case is the source of the frontier in §4.

**Example 1.** Take  $S = \{a, b\}$  with  $P_a = k^2$ ,  $P_b = k$ . Then  $\llbracket S, P \rrbracket W = \mathbf{Vec}(k^2, W) \oplus \mathbf{Vec}(k, W) \cong W^2 \oplus W \cong W^3$ , so  $\llbracket S, P \rrbracket \cong \text{Id}^3$ . Numerically  $\dim \llbracket S, P \rrbracket(k^d) = 3d$  for every  $d$  (verified computationally,  $d = 0, \dots, 4$ ). The two integers that produced this functor — the shape  $S$  and the partition  $3 = 2 + 1$  — have already vanished from the answer: only the total  $N = \sum_s n_s = 3$  remains. That disappearance is the subject of §2.

## 2 The collapse: $\llbracket - \rrbracket$ is neither full nor faithful over **Vec**, and not injective on objects

Over **Set** the extension functor  $\llbracket - \rrbracket : \mathbf{Fam}(\mathbf{Set}^{\text{op}}) \rightarrow [\mathbf{Set}, \mathbf{Set}]$  is **full and faithful** (Abbott–Altenkirch–Ghani): the container data  $(S, P)$  can be recovered from the functor, with  $S = \llbracket S, P \rrbracket(1)$  the value at a one-element set. Over **Vec** this recovery fails at its root.

*Remark 1* (the terminal slogan dies). The terminal object of **Vec** is the zero object  $0$ , so  $\llbracket S, P \rrbracket(0) = \bigoplus_s \mathbf{Vec}(P_s, 0) = 0$ : the shapes are invisible at the terminal object, and the **Set** slogan “ $F(1) = S$ ” has no analogue.

### 2.1 The finite collapse (headline; proved and lean-verified)

**Theorem 1** (finite collapse; **proved**). *Let  $S$  be finite and every  $n_s = \dim P_s$  finite, and put  $N := \sum_{s \in S} n_s = \dim(\bigoplus_s P_s)$ . Then there is a natural isomorphism  $\llbracket S, P \rrbracket \cong \text{Id}^N$ . Consequently:*

1.  $\llbracket S, P \rrbracket \cong \llbracket S', P' \rrbracket$  as functors **iff**  $N = N'$ ; a finite vector-space container is classified up to isomorphism of its extension by the single number  $N$ .
2. Since  $\text{End}(\text{Id}) = k$  is local and  $\text{End}(\text{Id}^N) = M_N(k)$  is semiperfect, Krull–Schmidt–Azumaya makes the decomposition into indecomposables unique; the recovered invariant is exactly the multiplicity  $N$ . The shape set  $S$  and the partition  $N = \sum_s n_s$  are not recoverable.

This is proved in `proofs/2026-08-18-linear-containers-vec.md` (Theorem 2.1; registry `linear-containers-vec.json`, node `part1-finite-collapse`, grade **proved**), with the co-Yoneda lemma  $\text{Nat}(h_P, G) \cong G(P)$  and the additivity of  $P \mapsto h_P$  as the engine.

**Theorem 2** (the explicit witness; **proved, lean-verified**). *The two containers*

$$p = (\{*\}, k^2) \quad \text{and} \quad p' = (\{1, 2\}, (k, k))$$

both have extension  $\text{Id}^2$ , i.e. both present the functor  $X \mapsto X \oplus X$ , yet they are **not isomorphic as containers**: their shape sets have different cardinalities ( $1 \neq 2$ ). Hence  $\llbracket - \rrbracket$  is not injective on objects.

Contrast **Set**: there the analogous containers  $(\{*\}, 2)$  and  $(\{1, 2\}, (1, 1))$  have *distinct* extensions  $X^2$  and  $2 \cdot X$ , separated already at  $X = 1$ . The collapse is precisely the failure of that separation. Theorem 2 is machine-checked; see §5.

## 2.2 Why: the hom formula and the failure of extensivity

The mechanism is visible one categorical level up, on hom-sets. Both computations below are proved in 2026-08-18-linear-containers-vec.md (Theorems 3.1–3.3; registry nodes `part2-hom-formula`, `part2-extensivity-crux`, both **proved**).

**Proposition 1** (hom formula; **proved**). *For vector-space containers  $(S, P)$ ,  $(T, Q)$ ,*

$$\begin{aligned} \text{Nat}(\llbracket S, P \rrbracket, \llbracket T, Q \rrbracket) &\cong \prod_{s \in S} \bigoplus_{t \in T} \mathbf{Vec}(Q_t, P_s), \\ \text{Fam}(\mathbf{Vec}^{\text{op}})((S, P), (T, Q)) &\cong \prod_{s \in S} \prod_{t \in T} \mathbf{Vec}(Q_t, P_s), \end{aligned}$$

where  $\coprod$  is the coproduct in **Set** (disjoint union): a container morphism assigns to each  $s$  a single choice  $t = f(s)$  together with a linear map in  $\mathbf{Vec}(Q_{f(s)}, P_s)$ .

**Theorem 3** (neither full nor faithful; **proved**). *The functor  $\llbracket - \rrbracket$  induces on each hom the canonical comparison*

$$\prod_s \prod_t \mathbf{Vec}(Q_t, P_s) \longrightarrow \prod_s \bigoplus_t \mathbf{Vec}(Q_t, P_s),$$

componentwise the map from the disjoint union  $\coprod_t$  into the biproduct  $\bigoplus_t$  sending a choice  $(t, \varphi)$  to the element supported at  $t$ . Over  $\mathbf{Vec}$  this map is neither surjective nor injective, so  $\llbracket - \rrbracket$  is **neither full nor faithful**.

1. Not full as soon as some shape  $s$  admits two shapes  $t \neq t'$  with  $\mathbf{Vec}(Q_t, P_s) \neq 0 \neq \mathbf{Vec}(Q_{t'}, P_s)$ : any natural transformation whose  $s$ -component is a genuine linear combination  $\varphi_t + \varphi_{t'}$  supported on two shapes is not in the image of a shape-function.
2. Not faithful as soon as  $S \neq \emptyset$  and  $|T| \geq 2$ . Over a field the zero linear map lies in every  $\mathbf{Vec}(Q_t, P_s)$ , so the two distinct container morphisms  $(u, 0)$  and  $(u', 0)$  — with  $u \neq u': S \rightarrow T$  and all position maps zero — are collapsed to the same (zero) natural transformation. The mechanism is exactly the biproduct:  $\bigoplus_t$  has a single shared zero across all summands, so it merges the  $|T|$  distinct disjoint-union-tagged zeros  $(t, 0)$  into that one zero element; the comparison is injective only away from the zero maps. Hence  $\llbracket - \rrbracket$  is **faithful iff**  $|T| \leq 1$  **or**  $S = \emptyset$ . By contrast **Set**'s coproduct is the disjoint union, which keeps a distinct tag on each zero, so the classical Abbott–Altenkirch–Ghani  $\llbracket - \rrbracket$  over **Set** remains faithful.

*Remark 2* (provenance of the faithfulness correction). The failure of faithfulness was surfaced by R. Langer's finite-field analogue (q-container Cor. 5.3, Ex. 5.4); over  $\mathbb{F}_q$  it is a finite point-count. (Draft note.)

The point is a single symbol. Over **Set** the same computation gives  $\text{Nat}(\llbracket S, P \rrbracket, \llbracket T, Q \rrbracket) \cong \coprod_s \bigoplus_t \mathbf{Set}(Q_t, P_s)$  where now  $\bigoplus_t = \coprod_t$  is the coproduct *in Set*, i.e. the disjoint union: container-hom equals Nat, and  $\llbracket - \rrbracket$  is full and faithful. **Set is extensive** — its coproduct is the disjoint union — and that is exactly what makes  $\llbracket - \rrbracket$  full. **Vec is not extensive** — its coproduct is a biproduct — and the fullness gap

$$\coprod_t \subsetneq \bigoplus_t$$

is the precise, quantified manifestation of that failure. The object-level collapse (Theorem 1) and the morphism-level fullness gap (Theorem 3) are *one* phenomenon: the failure of extensivity under the base change  $\mathbf{Set} \rightsquigarrow \mathbf{Vec}$ .

*Remark 3* (what survives; **proved**). The collapse used both finiteness of  $S$  and finite-dimensionality of positions; drop either and content returns. With infinite  $S$  and all  $P_s = k$ ,  $\llbracket S, P \rrbracket = \text{Id}^{\oplus |S|}$  is a genuine coproduct, not a product, of copies of Id; and for a single infinite-dimensional  $P$ ,  $h_P = \mathbf{Vec}(P, -)$  is an infinite-product functor, not built from Id under  $\oplus$ , and  $P$  is recovered from the representable (Lemma 1.2/1.3 of the source file). The finite recovery is sharp:  $\llbracket - \rrbracket$  determines the total space  $\bigoplus_s P_s$  up to isomorphism, and nothing about its partition into shapes (Corollary 3.4, **proved**).

### 3 Algebraic structure

Composition of vector-space containers is available on the finite-dimensional locus, and it too is shaped by the collapse.

**Proposition 2** (composition; **proved**). *For containers with finite-dimensional positions,*

$$\llbracket S, P \rrbracket \circ \llbracket T, Q \rrbracket \cong \llbracket S \times T, (P_s \otimes Q_t)_{(s,t)} \rrbracket,$$

*so on  $\text{Fam}(\mathbf{Vec}_{\text{fd}}^{\text{op}})$  the composition product is  $(S, P) \triangleleft (T, Q) = (S \times T, (P_s \otimes Q_t))$ , with unit  $I = (\{\ast\}, k)$  (extension Id).*

This is *not* the **Set** container composition, whose shapes form the dependent sum  $\coprod_{s \in S} T^{P_s}$ : linearity flattens the dependency (a linear map  $P_s \rightarrow \bigoplus_t V_t$  is a *sum of components*, not a *choice of branch*) into the plain product  $S \times T$ . Proved in 2026-08-18-linear-containers-vec.md, Prop. 4.1 (registry node **part3-composition**, **proved**).

The comonoids for this product identify a clean algebraic species.

**Theorem 4** ( $\triangleleft$ -comonoids are families of algebras; **proved**). *Let  $C = (S, P) \in \text{Fam}(\mathbf{Vec}_{\text{fd}}^{\text{op}})$  with  $S$  an arbitrary set and each  $P_s$  finite-dimensional. The  $\triangleleft$ -comonoid structures  $(\delta, \varepsilon)$  on  $C$  in  $(\text{Fam}(\mathbf{Vec}_{\text{fd}}^{\text{op}}), \triangleleft, I)$  are in natural bijection with families  $(A_s)_{s \in S}$  of unital associative  $k$ -algebra structures  $A_s = (P_s, \mu_s, \eta_s)$ , one on each position space. The bijection is  $\delta_{\text{shape}} = \Delta$  (the diagonal),  $\delta_s^\sharp = \mu_s$ ,  $\varepsilon_s^\sharp = \eta_s$ . On morphisms this upgrades to an isomorphism of categories*

$$\text{Comon}_{\triangleleft}(\text{Fam}(\mathbf{Vec}_{\text{fd}}^{\text{op}})) \cong \text{Fam}(\mathbf{Alg}_k^{\text{op}}),$$

*the cocommutative comonoids forming the full subcategory  $\text{Fam}(\mathbf{CAlg}_k^{\text{op}})$ .*

Proved in **proofs/2026-08-19-vec-comonoids-algebras.md** (registry node **part3-comonoid-algebras**, grade **proved**; hypotheses sharpened to finite-dimensional positions only,  $S$  arbitrary). The counit forces the shape comultiplication to the diagonal (the unique  $(\mathbf{Set}, \times)$ -comonoid on  $S$ ); contravariance turns  $\delta_s^\sharp$  into a multiplication  $P_s \otimes P_s \rightarrow P_s$  and  $\varepsilon_s^\sharp$  into a unit  $k \rightarrow P_s$ ; the comonoid laws become associativity and the two-sided unit law.

*Remark 4* (the algebroid guess, refuted in finite dimension; **proved**). Over **Set** the analogous fact is  $\triangleleft$ -comonoid  $\cong$  small category (Ahman–Uustalu; **Cont**  $\cong$  Fam(**Set**<sup>op</sup>)). One might hope  $\triangleleft$ -comonoid over **Vec**  $\cong$   $k$ -linear category (algebroid, Mitchell). This is *false* in finite dimension: because the counit forces  $\delta_{\text{shape}} = \Delta$ , the comultiplication never reaches an off-diagonal block  $P_a \otimes P_b$  ( $a \neq b$ ), so there is no hom-between-distinct-objects composition. The structure is a disjoint family of one-object  $k$ -linear categories =  $k$ -algebras. A genuine algebroid requires double-indexed positions and a matrix (bimodule) composition  $(P \circ Q)_{a,c} = \bigoplus_b P_{a,b} \otimes Q_{b,c}$  — the  $\bigoplus_b$  restoring the extensivity coproduct whose absence caused the strict collapse. That identification is classical (Bénabou; Mitchell) and is treated separately; the finite single-index result stands as above.

## Which monoidal / closed structures survive (high level)

Three companion results from the wider “containers over a base” programme locate the collapse inside the closed-structure theory. They are stated here at high level and *not* re-proved; each is proved in its own source file.

- **T1 (unit connectedness; proved)**. For a closed symmetric monoidal cocomplete base  $\mathcal{C}$ , the extension functor  $\llbracket - \rrbracket : \text{Fam}(\mathcal{C}^{\text{op}}) \rightarrow [\mathcal{C}, \mathcal{C}]$  is full and faithful *iff* the monoidal unit is *connected* ( $\mathcal{C}(I, -)$  preserves coproducts). Over **Vec** the unit is *disconnected* (the zero object makes  $|-| : \mathbf{Vec}_{\text{fd}} \rightarrow \mathbf{Set}$  fail to preserve coproducts), which is exactly why  $\llbracket - \rrbracket$  is not full — a base-general explanation of §2.
- **T2 (Dirichlet / Day tensor; proved)**. The pointwise Dirichlet tensor on Fam( $\mathcal{C}^{\text{op}}$ ) is Day convolution, and its closedness is controlled by a familial-representability condition that fails over both **Vec** and **Vec**<sub>fd</sub>.
- **T4-left (left closure; proved)**. The composition product  $\triangleleft$  is left-closed on the copower-tiny (collapse) locus, where  $p \triangleleft q = p \otimes q$ ; this is the locus on which Proposition 2 lives.

(These are internal results of MacBeth’s programme; grades are as recorded in memory and are cited, not re-derived, here.)

## 4 Composition and admissibility

Proposition 2 defines  $\triangleleft$  on *finite-dimensional* positions. The general question is whether Fam(**Vec**<sup>op</sup>) carries a composition product at all.

**Definition 3** ( $\triangleleft$ -admissibility). Fam( $\mathcal{C}^{\text{op}}$ ) is  $\triangleleft$ -*admissible* if the image of  $\llbracket - \rrbracket$  in  $[\mathcal{C}, \mathcal{C}]$  is closed under composition: for all  $p, q$  there is  $r$  with  $\llbracket r \rrbracket \cong \llbracket p \rrbracket \circ \llbracket q \rrbracket$ .

**Proposition 3** (admissibility is per-object; **proved**). Call  $P \in \mathcal{C}$  absorptive if  $X \mapsto [P, \llbracket q \rrbracket X]$  is an extension for every  $q$ . Then Fam( $\mathcal{C}^{\text{op}}$ ) is  $\triangleleft$ -admissible *iff* every object of  $\mathcal{C}$  is absorptive. There are two sources of absorptivity: copower-tiny (flexible) objects, where  $[P, \coprod_t Y_t] = \coprod_t [P, Y_t]$ ; and distributive/extensive (rigid) objects, via the **Set**-style mechanism.

Proposition 6.1 of proofs/2026-09-01-gap1-setxvec-proved.md (**proved**). In **Vec**<sub>fd</sub> every object is copower-tiny (finite-dimensional = finitely generated = copower-tiny; Lemma 5.1 there, **proved**), so Fam(**Vec**<sub>fd</sub><sup>op</sup>) is admissible on the collapse locus with  $\triangleleft = \otimes_k$  — this is Proposition 2 again.

#### 4.1 A proved positive result: $\mathbf{Set} \times \mathbf{Vec}_{\text{fd}}$ is admissible

**Theorem 5** (Gap 1 inhabited; **proved**). *On the finite-vec-support locus  $\mathcal{P} \subseteq \text{Fam}((\mathbf{Set} \times \mathbf{Vec}_{\text{fd}})^{\text{op}})$  (shape  $S$  and set-positions  $A_s$  arbitrary;  $\{s : V_s \neq 0\}$  finite), the base  $\mathcal{C} = \mathbf{Set} \times \mathbf{Vec}_{\text{fd}}$  is  $\triangleleft$ -admissible: for all  $p, q \in \mathcal{P}$  there is  $r = p \triangleleft q \in \mathcal{P}$  and an explicit natural isomorphism  $\Theta: \llbracket p \triangleleft q \rrbracket \Rightarrow \llbracket p \rrbracket \circ \llbracket q \rrbracket$  — both components built as maps (the  $\mathbf{Set}$ -component a bijection with explicit inverse, the  $\mathbf{Vec}$ -component a natural composite of hom-out-of-coproduct, tensor-hom, and finite-support isomorphisms), not matched merely by cardinality.*

Theorem 3.1 of `proofs/2026-09-01-gap1-setxvec-proved.md` (**proved**; the map was built and its bijectivity, invertibility, and naturality confirmed computationally with zero failures). The base  $\mathbf{Set} \times \mathbf{Vec}_{\text{fd}}$  is non-collapse, non-cartesian, and has a disconnected unit, so it exhibits admissibility *without* unit-connectedness: it factors a rigid (extensive)  $\mathbf{Set}$  part against a flexible (copower-tiny)  $\mathbf{Vec}$  part. One honesty caveat, itself proved: the product  $\triangleleft$  carries *no canonical* monoidal structure — the construction must choose a concentration slot — and the obstruction to canonicity is exactly the non-faithfulness of  $\llbracket - \rrbracket$  on objects (Theorem 2). A non-canonical monoidal structure does exist.

#### 4.2 The open frontier: is full $\mathbf{Vec}$ admissible? (Question (Q), OPEN)

For *finite-dimensional* positions everything works. For *full*  $\mathbf{Vec}$  the issue is that an infinite-dimensional position  $P$  is not copower-tiny, so  $[P, \bigoplus_t (-)]$  need not be an extension. By Proposition 3 the whole question reduces to a single test object. The sharpest one is  $E := k^{(\mathbb{N})}$  (countable-dimensional, free, non-tiny) against  $q = (\mathbb{N}, (k))$ , giving  $\llbracket q \rrbracket X = \bigoplus_{\mathbb{N}} X$  and

**Question 1** (the frontier; **OPEN**). Is

$$F(X) := \text{Hom}(E, \bigoplus_{\mathbb{N}} X) = \prod_{\mathbb{N}} \left( \bigoplus_{\mathbb{N}} X \right)$$

(row-finite  $\mathbb{N} \times \mathbb{N}$  matrices over  $X$ )

naturally isomorphic to a coproduct of representables  $\bigoplus_j \text{Hom}(N_j, -)$  in the additive functor category  $\text{Add}(\mathbf{Vec}, \mathbf{Vec})$ ? Equivalently: is  $F$  a *projective* object of  $\text{Add}(\mathbf{Vec}, \mathbf{Vec})$ ?

The stakes: **(Q) YES** would give an irreducible admissible base beyond Theorem 5; **(Q) NO** would confirm the “absorptive dichotomy” conjecture (every nice admissible base decomposes as rigid-extensive  $\times$  flexible-additive). *This is the one deep open problem in the theory and we state it as unsettled.*

What *is* proved is a precise homological reduction, all in ZFC, in `proofs/2026-09-02-vec-admissibility-rigidity.md` and `proofs/2026-09-04-Q-homological-reduction-and-target-dimension-dichotomy.md`:

- **Lemma R (Yoneda rigidity; proved)**. Every  $\alpha \in \text{Nat}(F, \text{id})$  has  $\alpha \circ \text{in}_n = 0$  for all but finitely many rows  $n$ . The engine is pure Yoneda:  $\prod_{\mathbb{N}} = h_E$  and  $\text{Nat}(h_E, \text{id}) = E = k^{(\mathbb{N})}$  is finite-support. This resurrects Baer–Specker-type rigidity *over a field*, where module slenderness fails.
- **Theorem E (homological reduction; proved, ZFC)**. With  $F_{\text{fin}} := \bigoplus_n A \subseteq F$  ( $A := \bigoplus_m \text{id}$ ) and  $Q := F/F_{\text{fin}}$ , for every  $M$  there is an exact sequence  $0 \rightarrow \text{coker}(\rho_M) \rightarrow \text{Ext}^1(Q, M) \rightarrow \text{Ext}^1(F, M) \rightarrow 0$ . Hence

$$F \text{ is projective} \iff \text{Ext}^1(F, M) = 0 \text{ for all } M.$$

So (Q) is exactly the single homological question “is  $\text{Ext}^1(F, -) \equiv 0$ ?”.

- **Theorem H (cardinality is neutral; proved, ZFC).** The id-dual  $D(h_N) = \text{Nat}(h_N, \text{id}) = N$  has *no*  $2^{\dim}$  blow-up (unlike  $\text{Hom}_{\mathbb{Z}}(\mathbb{Z}^{(\kappa)}, \mathbb{Z})$ ), so no cardinal invariant  $\dim F(X)$  or  $\dim \text{Nat}(F, M)$  separates  $F$  from a free functor. Both classical decision routes — single-target rigidity (all such  $\text{Nat}$ -groups vanish on  $Q$ ) and cardinality — are thereby *provably closed*; any resolution must engage  $\text{Ext}^1(F, -)$  at an infinite-dimensional target.

*Remark 5* (status of (Q); **OPEN**, conjectured independent of ZFC). Question 1 is **not settled**. Because both the cardinality engine (Theorem H) and the single-target engine are provably unavailable over a field, the residual is a Whitehead-type extension problem on the syzygies of  $F$ . The current *conjecture* — theorem-supported but *not* proved — is that (Q) is **independent of ZFC**: plausibly NO under  $2^{\aleph_0} < 2^{\aleph_1}$  and possibly YES under a failure of that hypothesis plus a uniformisation. Settling (Q) requires genuine set-theoretic input (Shelah; Eklof–Mekler transported to functor categories) and is the flagged frontier of the programme.

## 5 Formalisation

The object-level collapse (Theorem 2) is machine-checked in Lean 4 core, with *no* Mathlib dependency, in `lean/Containers/Containers/VecCollapse.lean`. Positions are carried by their finite basis types ( $k^2$  by `Bool`,  $k$  by `Unit`), and the linear extension is modelled as the  $\Pi$ -over-shapes  $\text{LExt}(S, \text{Pos}) X = (s : S) \rightarrow (\text{Pos } s \rightarrow X)$  — the single  $\Sigma \rightsquigarrow \Pi$  shift that *is* the biproduct collapse. The two containers  $p = (\{*\}, k^2)$  and  $p' = (\{1, 2\}, k)$  appear as `VecCollapse.p` and `VecCollapse.p'`. The theorem `Containers.VecCollapse.collapseIso` exhibits a full natural isomorphism `LinNatIso p p'` (both directions, both triangle identities, and naturality in  $X$ ), while `Containers.VecCollapse.shapes_not_equiv` certifies (`sorry-free`, and axiom-free — `#print axioms` reports `[]`) that the shape sets `Unit` and `Bool` admit no bijection. Together they are the machine-checked witness that  $\llbracket - \rrbracket$  presents the same functor  $X \mapsto X \oplus X$  by two non-isomorphic containers — Theorem D(ii), **lean-verified**. The isomorphism `collapseIso` depends only on `Quot.sound` (function extensionality).

## 6 Summary and outlook

| Statement                                                                                                             | Source                        | Grade                                |
|-----------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------------------------------|
| Extension $\llbracket S, P \rrbracket W = \bigoplus_s \mathbf{Vec}(P_s, W)$ ; $\Sigma \rightarrow \Pi$ biproduct form | 2026-08-18, Def. 0.2          | definition                           |
| Finite collapse $\llbracket S, P \rrbracket \cong \text{Id}^N$ ; classified by $N$ alone (Thm 1)                      | 2026-08-18, Thm 2.1           | <b>proved</b>                        |
| Witness $(\{*\}, k^2) \not\cong (\{1, 2\}, k)$ , same functor $X \mapsto X \oplus X$ (Thm 2)                          | 2026-08-18 / VecCollapse.lean | <b>proved, verified</b> <b>lean-</b> |
| $\llbracket - \rrbracket$ neither full nor faithful; faithful iff $ T  \leq 1$ or $S = \emptyset$ (Thm 3)             | 2026-08-18, Thm 3.3           | <b>proved</b>                        |
| Composition $(S, P) \triangleleft (T, Q) = (S \times T, (P_s \otimes Q_t))$ (Prop 2)                                  | 2026-08-18, Prop 4.1          | <b>proved</b>                        |
| $\triangleleft$ -comonoids $\cong$ families of $k$ -algebras (Thm 4)                                                  | 2026-08-19, Thm 7             | <b>proved</b>                        |
| Admissibility $\Leftrightarrow$ every object absorptive (Prop 3)                                                      | 2026-09-01, Prop 6.1          | <b>proved</b>                        |
| $\mathbf{Set} \times \mathbf{Vec}_{\text{fd}}$ is $\triangleleft$ -admissible (Thm 5)                                 | 2026-09-01, Thm 3.1           | <b>proved</b>                        |
| Homological reduction: $F \text{ proj.} \Leftrightarrow \text{Ext}^1(F, -) \equiv 0$ (Thm E)                          | 2026-09-04, Thm E             | <b>proved</b> (ZFC)                  |
| No cardinal invariant separates $F$ from free (Thm H)                                                                 | 2026-09-04, Thm H             | <b>proved</b> (ZFC)                  |
| <b>(Q)</b> Is full $\mathbf{Vec}$ admissible / $F$ projective? (Question 1)                                           | 2026-09-02/04                 | <b>OPEN</b> (conj. indep. ZFC)       |

**Outlook.** The theory of vector-space containers is complete and clean on the finite-dimensional locus: the extension functor is neither full nor faithful (faithful only when the target has  $\leq 1$  shape or the source is empty) and is not injective on objects, the failure is exactly the non-extensivity of  $\mathbf{Vec}$ , comonoids are families of algebras, and  $\mathbf{Set} \times \mathbf{Vec}_{\text{fd}}$  realises admissibility off the connected-unit pole. The single deep frontier is Question (Q) — whether *infinite-dimensional* linear resources can still absorb external shape — now sharpened to a precise, ZFC-reduced homological question ( $\text{Ext}^1(F, -) \equiv 0$ ?) with both classical decision routes provably closed and independence of ZFC conjectured. Settling it is the natural next step and would either produce a genuinely new irreducible composition base or complete the absorptive-dichotomy classification.

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*Draft — work in progress. Prepared for Neil Ghani / external requester, 2026-09-05. Internal sources: proofs/2026-08-18-linear-containers-vec.md, proofs/2026-08-19-vec-comonoids-algebras.md, proofs/2026-09-01-gap1-setxvec-proved.md, proofs/2026-09-02-vec-admissibility-rigidity.md, proofs/2026-09-04-Q-homological-reduction-and-target-dimension-dichotomy.md, lean/Containers/Containers/VecCollapse.lean. Grades per the corresponding registry files.*